



Radio Science

RESEARCH ARTICLE

10.1002/2015RS005910

Special Section:

Ionospheric Effects Symposium
2015

Key Points:

- Combined ground- and space-based measurements of ionosphere signatures associated with natural hazards are presented
- JPL-modeled and measured TEC perturbations are in good agreement
- Space-based platforms are expected to play an integral part of future early warning systems

Correspondence to:

A. Komjathy,
Attila.Komjathy@jpl.nasa.gov

Citation:

Komjathy, A., Y.-M. Yang, X. Meng, O. Verkhoglyadova, A. J. Mannucci, and R. B. Langley (2016), Review and perspectives: Understanding natural-hazards-generated ionospheric perturbations using GPS measurements and coupled modeling, *Radio Sci.*, 51, 951–961, doi:10.1002/2015RS005910.

Received 1 DEC 2015

Accepted 15 JUN 2016

Accepted article online 20 JUN 2016

Published online 13 JUL 2016

Review and perspectives: Understanding natural-hazards-generated ionospheric perturbations using GPS measurements and coupled modeling

Attila Komjathy¹, Yu-Ming Yang¹, Xing Meng¹, Olga Verkhoglyadova¹, Anthony J. Mannucci¹, and Richard B. Langley²
¹Jet Propulsion Laboratory, California Institute of Technology, Pasadena, California, USA, ²Department of Geodesy and Geomatics Engineering, University of New Brunswick, Fredericton, New Brunswick, Canada

Abstract Natural hazards including earthquakes, volcanic eruptions, and tsunamis have been significant threats to humans throughout recorded history. Global navigation satellite systems (GNSS; including the Global Positioning System (GPS)) receivers have become primary sensors to measure signatures associated with natural hazards. These signatures typically include GPS-derived seismic deformation measurements, coseismic vertical displacements, and real-time GPS-derived ocean buoy positioning estimates. Another way to use GPS observables is to compute the ionospheric total electron content (TEC) to measure, model, and monitor postseismic ionospheric disturbances caused by, e.g., earthquakes, volcanic eruptions, and tsunamis. In this paper, we review research progress at the Jet Propulsion Laboratory and elsewhere using examples of ground-based and spaceborne observation of natural hazards that generated TEC perturbations. We present results for state-of-the-art imaging using ground-based and spaceborne ionospheric measurements and coupled atmosphere-ionosphere modeling of ionospheric TEC perturbations. We also report advancements and chart future directions in modeling and inversion techniques to estimate tsunami wave heights and ground surface displacements using TEC measurements and error estimates. Our initial retrievals strongly suggest that both ground-based and spaceborne GPS remote sensing techniques could play a critical role in detection and imaging of the upper atmosphere signatures of natural hazards including earthquakes and tsunamis. We found that combining ground-based and spaceborne measurements may be crucial in estimating critical geophysical parameters such as tsunami wave heights and ground surface displacements using TEC observations. The GNSS-based remote sensing of natural-hazard-induced ionospheric disturbances could be applied to and used in operational tsunami and earthquake early warning systems.

1. Introduction

Earthquakes, tsunamis, and volcanoes are among the most disruptive forces faced by humankind, often endangering heavily populated coastal areas. Ground-based receivers using satellite signals from global navigation satellite systems (GNSS) such as the Global Positioning System (GPS) have served as primary sensors for over a decade to measure signatures associated with such natural hazards. These signals are seismic deformation measurements, coseismic ground vertical displacements, and even real-time ocean buoy positioning estimates. Researchers have also begun using GPS observables to compute the ionospheric total electron content (TEC) to measure, image, and monitor postseismic ionospheric disturbances caused by various natural hazards such as earthquakes, tsunamis, and asteroid impacts [e.g., Calais and Minster, 1995; Ducic et al., 2003; Mazzotti et al., 2005; Bergeot et al., 2009; Astafyeva and Heki, 2009; Astafyeva et al., 2009, 2011, 2013a, 2013b, 2014; Yang et al., 2014a; Komjathy et al., 2015a; Jin et al., 2015]. Volcanic eruptions also generate TEC perturbations triggered by acoustic and gravity waves [e.g., Calais and Minster, 1998; Dautermann et al., 2009a, 2009b; Heki, 2006; Zabolotn et al., 2016; Jonah et al., 2016]. For example, acoustic wave signatures in TEC perturbations is becoming increasingly important because of the paucity of data from other means, such as infrasound data, to measure energy content of volcanic eruptions [e.g., Heki, 2006; Watada and Kanamori, 2010; Nakashima et al., 2014]. Lay et al. [2013] examined thunderstorms generating anomalous TEC perturbations with the largest storm-related TEC perturbations of ~ 1.4 TECU (TECU, $1 \text{ TECU} = 10^{16} \text{ el m}^{-2}$) over a typical nighttime background value of several TECUs.

Earthquakes are known to generate electron density irregularities in the ionosphere and TEC variations through atmospheric waves [Yuen et al., 1969; Kelley et al., 1985; Calais and Minster, 1995, 1998; Artru et al., 2001;

Galvan et al., 2011, 2012; *Rolland et al.*, 2011, 2013; *Komjathy et al.*, 2015b]. There are three types of atmospheric waves generated by earthquakes: (1) direct acoustic waves generated near the epicenter, (2) gravity waves generated by an associated tsunami, and (3) secondary acoustic waves excited by the Rayleigh surface waves outside the epicenter area [*Heki*, 2006]. Although ground-based GPS receivers are limited to landmasses and islands, the GPS measurements are well suited to monitor ionospheric activity associated with earthquakes within about 1500 km of a ground-based GPS receiver to infer key properties such as velocity, direction, and magnitude of traveling ionospheric disturbances (TIDs).

Tsunamis generated by earthquakes often endanger heavily populated coastal areas [e.g., *Makela et al.*, 2011; *Heki*, 2011; *Arrowsmith et al.*, 2012; *Yang et al.*, 2012, 2014b; *Astafyeva et al.*, 2014; *Roy et al.*, 2014; *Jin et al.*, 2015]. The tsunami of 11 March 2011, following the Tohoku-Oki earthquake in Japan, caused over 20,000 fatalities. The total number of deaths from the tsunami following the 2004 Sumatra earthquake reached over 230,000 [e.g., *Ochipinti et al.*, 2013]. These, along with the Haida Gwaii (formerly known as the Queen Charlotte Islands) event [*United States Geological Survey*, 2016], highlight the limits to emergency managers' ability to deploy effective ground-based detection systems, partially due to significant flaws in our understanding of the coupling of earthquake and tsunami hazards [*Song et al.*, 2008; *Song and Han*, 2011]. The shortage of robust, cost-effective observations with extensive basin-wide coverage restricts our ability to properly characterize events and accurately predict their impacts. There remains much to learn about the characteristics of these interactions between the Earth's surface and ionosphere.

Recent advances at NASA's Jet Propulsion Laboratory using ground-based and spaceborne GPS measurements resulted in new and innovative GPS applications including the use of ionospheric measurements to detect small fluctuations in GPS signals between the spacecraft and GPS receivers caused by natural hazards occurring on or near the Earth's surface. This continuing research is expected to provide assistance for early warning systems for tsunamis, earthquakes, volcanic eruptions, and asteroid atmospheric impacts using real-time data from GPS and other GNSS.

In this paper we review and discuss examples of recent natural hazards that generated TEC perturbations. We elaborate on the state-of-the-art of retrievals of tsunami wave heights and ground surface displacements using TEC observations. We present results for imaging and coupled lower-atmosphere-thermosphere-ionosphere modeling of ionospheric TEC perturbations.

By studying the propagation properties of ionospheric perturbations generated by natural hazards along with applying sophisticated first-principles physics-based ionospheric modeling, the seismic-ionospheric community is on track to develop new ground- and space-based technologies that can potentially save human lives and minimize property damage.

2. Background and Motivation

Many fatalities occur when small earthquakes followed by large tsunamis go undetected. Accuracy in determination of the tsunami scale—a quantitative representation of the tsunami potential destructive force—is critical. It serves as a threshold for issuing a tsunami warning to affected areas [e.g., *Song*, 2007; *Song et al.*, 2012]. Earthquake source modeling can be improved using joint inversions of land-based estimates of coseismic deformation from GPS, near-shore wave gauges, along with novel GPS ionosphere-derived tsunami measurements. The results include better earthquake source estimation, initialization of tsunami propagation in a near-source coastline, and improvement in inundation forecasts by a factor of two as the increase in earthquake magnitude is proportional to the square of the propagation speed [*Melgar et al.*, 2013; *Melgar and Bock*, 2013, 2015]. Understanding the causal relationship between seismic waves and tsunami generation is key to developing future systems capable of increasing warning time and reducing fatalities as well as material damage.

Determining earthquake moment magnitudes (M_w) using ground-based measurements leads to large uncertainties [*Tang et al.*, 2012]. Due to the complexities of interpreting seismometer measurements and epistemic uncertainties in critical parameters such as Earth structure, the uncertainties in M_w may persist for months and years after the events, such as M_w 8.8 to 9.2 estimates for the Tohoku-Oki event and M_w 8.3 ± 3.3 for the 1964 Alaska earthquake [*Kanamori*, 1977; *Crowell et al.*, 2013]. Tsunamis are predominantly generated by earthquakes (83%). Earthquake-induced landslides (6%) and volcanic eruptions are other sources of tsunamis. Seismic data have been the primary measurements for tsunami studies. For earthquake-generated

tsunamis, the earthquake magnitude is related to the tsunami energy but the relationship is not yet well understood. As an example, a 0.2 increase in earthquake magnitude (by definition, defined in log scale) could correspond to a doubling of tsunami amplitude [Titov *et al.*, 2011]. The Tohoku-Oki tsunami energy (3×10^{15} J) was about 0.1% of the seismic radiation energy (2.84×10^{18} J) [Song *et al.*, 2012]. The substantially different earthquake magnitudes derived from this seismic data, ranging from M_w 7.9 to 9.2, reflect gaps in our understanding [Simons *et al.*, 2011]. The reasons for the uncertainties include the high-frequency nature of the 3-D seismic waves and the propagation effects due to attenuation and scattering [Tang *et al.*, 2012].

Scientists have recently come to the understanding that seismic data alone are unable to provide the level of accuracy required for tsunami modeling and energy transfer estimates [Melgar and Bock, 2013, 2015]. Other ground-based techniques exist to estimate tsunami energy including the use of coastal GPS data [Song, 2007] and tsunameters [Tang *et al.*, 2012]. Atmospheric and ionospheric disturbances are promptly generated by earthquakes and tsunamis and contain important geophysical signals on earthquake-tsunami coupling. However, there exists no other measurement system that is capable of substantially contributing to simultaneously estimating both seismic radiation energy and tsunami energy using spaceborne measurements. The combination of ground-based and spaceborne measurements provides a unique opportunity to study the spatial extent of seismic events while reducing systematic errors in simultaneously estimated seismic and tsunami energies. Other ground-based measurement techniques contribute to estimating either seismic or tsunami energies but not both.

Recent advances in GPS data processing have demonstrated that ground-based GPS receivers alone are capable of measuring TEC perturbations generated by acoustic and gravity waves in real time; however, the ground-based measurements are limited in spatial extent [e.g., Komjathy *et al.*, 2013a, 2013b, 2015a, 2015b]. The Tohoku-Oki earthquake and tsunami data processing results have demonstrated that Rayleigh surface waves are observed in the ionosphere followed by the acoustic-wave-generated perturbations. Minutes later, the gravity-wave-derived TEC perturbations are visible after the earthquake [e.g., Galvan *et al.*, 2011, 2012]. Surface Rayleigh waves are an important seismic signature and there is a direct link through dynamic coupling between the energy of ground displacements and the Rayleigh waves [Lognonné *et al.*, 2006a, 2006b; Rolland *et al.*, 2011].

In the following sections, we show examples of TEC perturbations generated by the Tohoku-Oki event using ground-based and spaceborne observations. We review observations, modeling results, and envision what the next steps should be to use TEC observations for retrieving tsunami wave height and ground surface displacements to help scientists better understand the energy transfer between earthquakes and tsunamis.

3. Method of Deriving TEC Measurements

The ionosphere is a dispersive medium for GNSS signals. The refractive index for a radio wave of frequency f (to the first order) is $n_p \approx 1 - 40.3N_e/f^2$, where n_p is the phase refractive index and N_e is the electron density [e.g., Komjathy, 1997, 2014]. As a radio wave travels through the ionosphere, one can calculate the excess phase delay $\delta\tau_p$ (in meters) from the refractive index n_p as

$$\delta\tau_p = -40.3 \cdot \text{TEC} / f^2. \quad (1)$$

GPS measurements from, e.g., Gravity Recovery and Climate Experiment (GRACE) dual-satellite precise orbit determination receivers can be used to calculate GPS-derived TEC time series from the carrier phase (Φ_1 and Φ_2) and pseudorange data (P_1 and P_2) using the L1 and L2 GPS frequencies. Furthermore, the GPS-derived TEC is given as

$$\text{TEC} = \frac{f_1^2 f_2^2}{40.3(f_1^2 - f_2^2)} (B - (\Phi_2 - \Phi_1)), \quad (2)$$

where f_1 and f_2 are the GPS L1 and L2 frequencies, Φ is the carrier phase measurement, and B is the ambiguity term defined in Mannucci *et al.* [1998] and Stephens *et al.* [2011]. In addition, this approach also applies to the retrieval of the change of the TEC time series $\Delta\text{TEC}(t)$ from GRACE K/Ka band ranging signals $dR(t)$, which is defined as

$$\Delta\text{TEC}(t) = dR(t) \cdot f_{ka}^2 / 40.3, \quad (3)$$

where $f_{ka} = 32$ GHz.

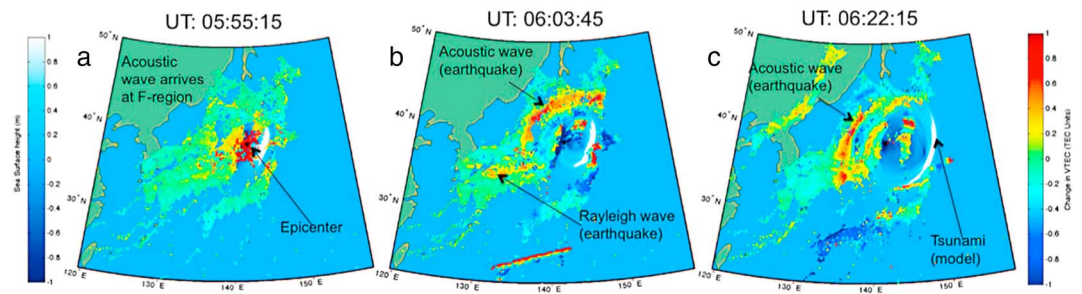


Figure 1. GPS is a powerful tool for imaging earthquakes and tsunamis. Ground-based GPS ionospheric data will show earthquake-generated (b) Rayleigh, (a and b) acoustic, and (c) gravity waves in the ionosphere. However, for studying smaller yet equally devastating events, we will need to add spaceborne observations (see Animation 1, 2016, <https://www.youtube.com/watch?v=iR6ZfG1gAms>, accessed on 23 April 2016). Figure based on Galvan *et al.* [2012].

4. Natural Hazard Investigation and Analysis Using GPS Observations

In this section, we summarize JPL processing results for the 2011 Tohoku-Oki earthquake and tsunami using ground-based and spaceborne observations. This is the first time ground-based and spaceborne measurements are discussed together in the context of a specific natural hazard event to emphasize the advantages of combining the two types of measurements.

The use of ground-based observations. On 11 March 2011, a large ($M_w = 9.0$) earthquake occurred off the Pacific coast of Tohoku, Japan (38.103°N , 142.861°E), at 05:46:23 UTC. It triggered a powerful tsunami with destructive runup waves reaching heights of up to 10 m. Concurrent GPS-based remote sensing results indicate that the coseismic TIDs may be characterized with horizontal propagation speeds of 3 km/s, 1 km/s, and 200 m/s that were found to be in agreement with the propagation speeds of Rayleigh waves (3.3 km/s), acoustic waves (1 km/s at an altitude of 300 km in the ionosphere), and gravity waves triggered by a fault slip and its following tsunami (200 m/s), respectively [Galvan *et al.*, 2012]. It has been demonstrated that large and dense GPS networks provide new opportunities to detect ionospheric signatures of large earthquakes such as the Tohoku-Oki earthquake and the resulting tsunami-induced ionospheric responses. However, TEC measurements from GPS satellites to ground-based receivers are often affected by measurement biases due to “phase cancelation effects.” Phase cancelation effects, prominent in integrated measurements of signals from satellites (such as TEC), were introduced by Georges and Hooke [1970].

In Figure 1, we illustrate the application of ground-based GPS measurements of TEC perturbations generated by a large tsunami. In the panels of Figure 1, we show about 1200 GPS receivers as part of the Japanese GEONET array imaging the Tohoku-Oki tsunami on 11 March 2011, using three subsequent epochs. GPS-derived TEC observations are able to resolve the tsunami-generated gravity waves impacting the ionosphere and causing TEC perturbations at the 1 TECU level (TEC range of ± 1 TECU on the right color bar). The panels also demonstrate that the JPL tsunami model predicted tsunami waves (changes in tsunami waves up to ± 1 m on the left color bar) [e.g., Song, 2007; Song and Han, 2011]. The ionospheric images of the tsunami waves are consistent with the modeling results. The MOST (for “Method of Splitting Tsunami”) model [Titov and Gonzales, 1997] for simulating tsunami waves also appears to be consistent in time and space with waves measured by GPS TEC and modeled by the JPL tsunami model [Galvan *et al.*, 2011]. Propagation speeds can distinguish the Rayleigh surface waves, acoustic waves, and gravity waves. However, it is important to point out that the Tohoku-Oki tsunami tragically turned out to be a “perfect storm.” The large earthquake generated an enormous tsunami during daytime when the ionosphere was fully developed. However, the impact zone was instrumented with 1200 survey-quality, high-precision GPS receivers continuously collecting GPS data during the event, which is not always the case. GPS absolute TEC products have an uncertainty of several TECU with an expected tsunami-generated ionospheric signature at the 1 TECU level. For successfully inverting measurements with tsunami amplitudes smaller than about 40 cm, we find that we will need to use all available data sources, including spaceborne high-precision GPS measurements, to reduce uncertainties in coupling between earthquakes and tsunamis [Komjathy *et al.*, 2012, 2014].

To help decrease the uncertainties in estimating earthquake moment, we use ground-based GPS ionospheric TEC measurements that we have acquired over the years. One of our research goals has been to demonstrate

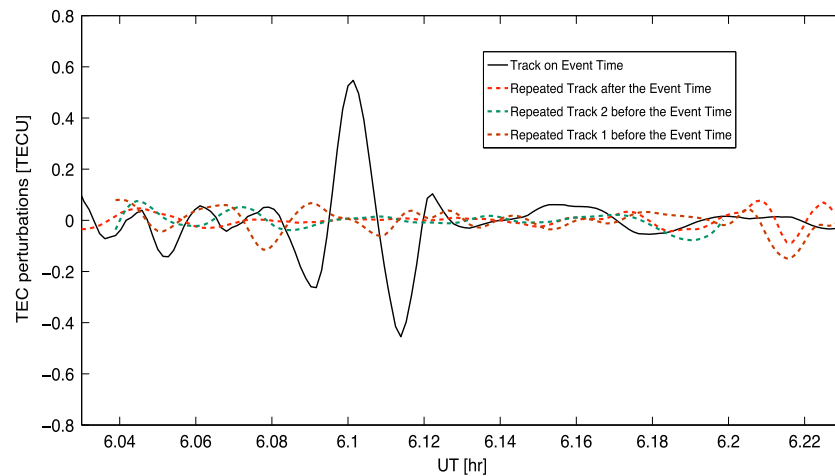


Figure 2. The TEC perturbations (black line) that correspond to different GRACE tracks passing through the region close to the event time. Significant TEC disturbances are observed about 20 min after the Tohoku-Oki main shock in Alaska. Other TEC waveforms (dashed lines) present the observations corresponding to repeated tracks traveling the same area (Alaska) at the same UTC on days before and after the event day. Results indicate that no significant TEC perturbations were observed 1 day before and after the track corresponding to the event time.

that tsunami wave amplitudes and propagation speeds observed by TID measurements can be inverted using a physics-based model.

The use of spaceborne observations. To better understand the earthquake to tsunami energy transfer and its source along with increasing the spatial and temporal coverage over the oceans, a successful inversion will require the use of spaceborne measurements. To take advantage of the growing number of low Earth orbiters including the current Constellation Observing System for Meteorology, Ionosphere, and Climate (COSMIC at CDAAC (COSMIC Data Analysis and Archive Center) (2016, <http://cdaac-www.cosmic.ucar.edu/cdaac/products.html>, accessed on 23 April 2016, hereinafter referred to as CDAAC, online report, 2016)), upcoming COSMIC-2 constellation, additional NASA remote sensing spacecraft such as GRACE [Physical Oceanography Distributed Archive Center, 2016], and *e-POP/CASSIOPE* [2016], we now routinely process and analyze the ionospheric radio occultation and airglow measurements for investigating earthquake- and tsunami-generated TEC perturbations.

The GRACE intersatellite precise range measurements provide a new opportunity to analyze the interaction of atmospheric waves and ionospheric disturbances associated with the 2011 Tohoku-Oki earthquake. Here we report on ionospheric TEC derived from measurements made from instruments on board the GRACE spacecraft. We examined and compared regional seismic measurements, infrasound signals, ground-based GPS network measurements, and GRACE Level 1-B observations a day before and after the earthquake event to detect coseismic ionosphere-thermosphere perturbations. At the time of the Tohoku-Oki earthquake on 11 March 2011, the twin spacecraft were orbiting at an altitude of ~ 450 km over Alaska. Significant TEC fluctuations (up to 0.6 TEC units, as shown in Figure 2) were observed ~ 8 min after the arrival of seismic and infrasound waves on the ground in Alaska, ~ 20 min after the Tohoku-Oki main shock at 05:46:23 UTC. The results of three-dimensional ionosphere-thermosphere modeling and infrasound ray tracing simulations are consistent with the arrival time and physical characteristics of the disturbances at GRACE. This is the first time that ionospheric disturbances associated with an earthquake are clearly attributable to perturbations at such high altitudes [Yang *et al.*, 2014b]. Other research by Coisson *et al.* [2015] investigated ionospheric radio occultation measurements to detect tsunami-generated TEC perturbations using COSMIC measurements alone.

In summary, increasing the temporal and spatial coverage over the oceans using the combination of ground-based GPS and spaceborne ionospheric measurements will help quantify our understanding of the relationship between earthquakes and the subsequent tsunamis while making it possible to quantify energies for smaller and therefore more frequent events. Such capability is imperative to understand and quantify the

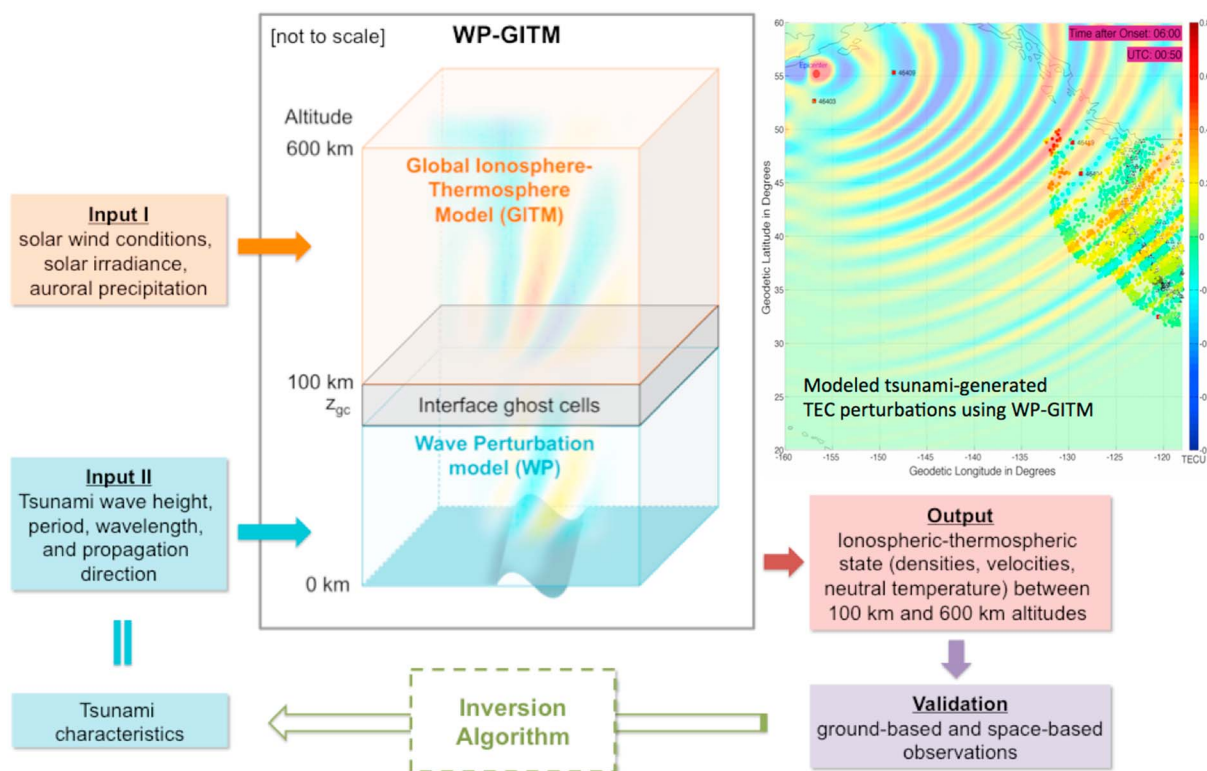


Figure 3. The JPL Wave Propagation-Global Ionosphere-Thermosphere Model (WP-GITM) flowchart.

wave coupling of earthquakes and tsunamis to the ionosphere. This will help scientists gain a better understanding of errors affecting the tsunami wave heights and vertical surface displacements that have direct impact on the mechanism coupling the surface to the ionosphere.

5. Modeling of Earthquake- and Tsunami-Ionosphere Coupling

Physics-based modeling is paramount in interpreting observations and reproducing tsunami-gravity-wave-ionosphere interactions. The earliest modeling work can be traced back to a two-dimensional (2-D) analytical model developed by *Peltier and Hines* [1976]. Their result corroborates the growth of the gravity wave amplitude with height. Later, Peltier and Hines's model was adopted by *Artru et al.* [2005a, 2005b] to confirm the observed travel time of a gravity wave propagating from the ocean surface to the ionosphere. More advanced models, including three-dimensional (3-D) time-dependent numerical models [*Occhipinti et al.*, 2006, 2008, 2011; *Vadas and Nicolls*, 2012; *Mai and Kiang*, 2009] and a spectral full-wave model [*Hickey et al.*, 2009], have been developed and applied to simulate ionospheric perturbations caused by tsunamis.

All models mentioned above contribute significantly to the understanding of tsunami-ionosphere coupling. However, these models either rely on an empirical atmospheric and ionospheric background that lacks self-consistent coupling between neutrals and ions or neglect some physical processes (such as chemical reactions, heat conduction, and radiative cooling) and space weather effects in the upper atmosphere, which limits the modeling ability to represent reality.

To overcome the limitations of previous models and better capture the upper atmospheric dynamics affected by tsunami-driven gravity waves, JPL developed the WP-GITM (Wave Perturbation-Global Ionosphere-Thermosphere Model), a 3-D tsunami-ionosphere coupling model, by implementing tsunami-induced gravity waves into the first-principles model of the ionosphere and thermosphere, GITM [*Ridley et al.*, 2006]. The WP-GITM model is described in detail by *Meng et al.* [2015]. As shown in Figure 3, the inputs of WP-GITM

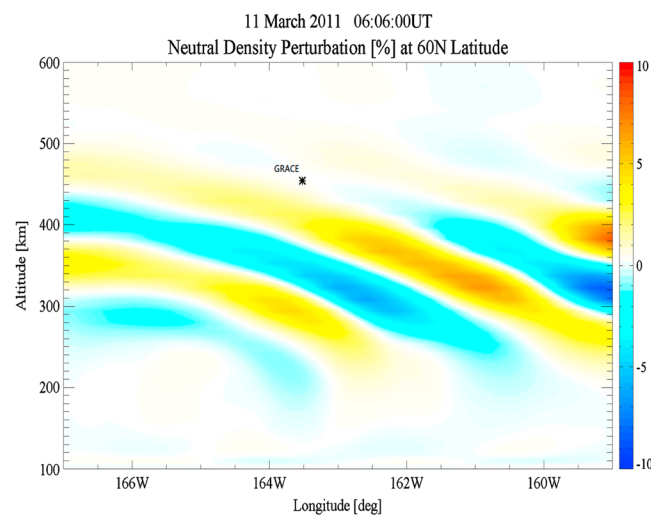


Figure 4. Snapshot of GITM-simulated neutral density perturbation 20 min after the main shock with approximate GRACE location (denoted by asterisk).

include tsunami characteristics: the tsunami wave height, wave period, wavelength, and propagation direction along with the standard inputs for externally driven GITM including but not limited to solar wind conditions and interplanetary magnetic field conditions. The output of WP-GITM is given by the GITM solution: the upper atmospheric state between 100 km and 600 km altitudes. For validation, the simulated TEC perturbations are routinely compared against the TEC perturbations derived from GPS observations. As an example, in the upper right panel of Figure 3, we show a WP-GITM simulation for the 1964 M_w 9.2 Alaska earthquake and ensuing tsunami event generating TEC perturbations with those reaching the West Coast of the U.S. (for complete

model simulations see Animation 2 (<https://www.youtube.com/watch?v=3m0l5bmA8-k>, accessed on 23 April 2016).

Other researchers compared TEC results with the state-of-the-art Embry-Riddle Aeronautical University (ERAU) model [Snively, 2013; Heale and Snively, 2015; Zettergren and Snively, 2015; Hickey *et al.*, 2015], and they found a good agreement between the ERAU model and acoustic-wave-generated TEC perturbations.

In our research, we applied the WP-GITM model to simulate the ionospheric perturbations after the Tohoku-Oki earthquake. The simulation domain covers 50°N to 66°N in geodetic latitude, 159°W to 167°W in geodetic longitude, and up to 600 km in altitude. To model the earthquake-generated perturbations, we perturbed the simulated east wind at the lower boundary of GITM starting at the arrival of infrasound and seismic waves (05:57:30 UTC) in the region. The perturbation is a cosine wave with the duration of 90 s, a period of 1 min and 200 m/s zonal wind perturbation amplitude corresponding to 0.02 Pa at 100 km altitude. These parameters were obtained from the coseismic infrasound waves observed by the infrasound station in Alaska (designated as IMS I53US). The simulation results support the conclusion that infrasound waves take about 8 min to propagate from the Earth's surface to the height of the GRACE satellites. They generate up to 4% amplitude density perturbations (against the ambient air density background) as shown in Figure 4. The results of JPL's 3-D atmosphere-ionosphere modeling are consistent with the arrival time and physical characteristics of the disturbances at GRACE [Yang *et al.*, 2014b].

JPL has begun working on an inversion algorithm using ground-based and spaceborne GPS observations collected near the locations of seismic events. The high-precision TEC data can be edited, noise reduced, and band-pass filtered using JPL's Pylono software suite [Komjathy *et al.*, 2013b]. Simultaneously, modeled TEC perturbations are generated using JPL's WP-GITM model. The model inversion is carried out using the Runge-Kutta iterative approach to estimate tsunami wave heights matching the observed TEC perturbations. The final product estimates of tsunami wave heights or vertical ground displacement over land allow us to compute tsunami source energy and improved tsunami scales (i.e., potential destructive force) [Komjathy *et al.*, 2015a]. Another technique presented by Rakoro *et al.* [2015] used tsunami normal modes in the atmosphere coupled with the ionosphere. They used simulated ground-based TEC data to retrieve tsunami wave heights reporting a few centimeter level precision for tsunami wave height retrievals.

We performed initial model simulations using GPS alone and combined GPS and spaceborne-data-derived surface displacements as a function of observed TEC perturbations in TECU [Komjathy *et al.*, 2014]. Since GPS absolute TEC products have an uncertainty of about 2 to 3 TECU, the JPL model inversion simulations indicate a 20–40 cm uncertainty in the GPS-ionospheric-measurement-predicted tsunami wave heights with wave amplitudes of less than 30 cm. On the other hand, the inclusion of potential spaceborne measurements,

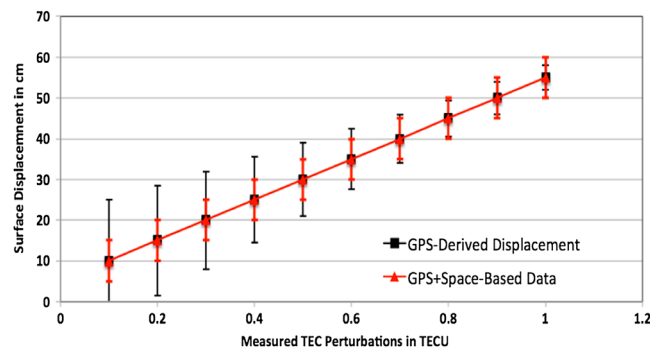


Figure 5. Illustration of ground-based and spaceborne-data-derived surface displacements and uncertainties.

tions in TECU. Figure 5 indicates that when including spaceborne measurements with, e.g., COSMIC data constraining the model solution, uncertainties of parameter estimates will decrease to a 5 cm level tsunami wave height (red error bars).

Furthermore, we continue investigating the various error sources affecting the spaceborne measurements and how the temporal and spatial variability of the different data sources will impact the error estimates of tsunami wave heights and vertical surface displacements. Each data set represents a unique set of challenges, with differing data noises and occasional inconsistencies among measurements. Quality control of the data sets is paramount for using the data sets to achieve centimeter level retrieval accuracies. We work on evaluating the accuracies of tsunami wave heights and ground vertical displacement estimates using ground truth measurements collected by the Deep-ocean Assessment and Reporting of Tsunamis [Deep-ocean Assessment and Reporting of Tsunamis, 2016] buoys (tsunami wave heights) and GPS-derived high-rate precise point positioning solutions. It remains crucial to understand which spaceborne measurement types contribute to the successful inversion the most (i.e., smallest error bar). We anticipate that spaceborne data types will become an important part of future natural hazards warning systems potentially helping save human lives and reducing the level of economic damage.

Our efforts tie directly into the initiative by the Global Geodetic Observing System (GGOS) of the International Association of Geodesy in coordinating research teams and representatives of national agencies participating in an international working group for implementation of a Global Navigation Satellite System Augmentation to Tsunami Early Warning Systems (GATEW) (Global Geodetic Observing System—Tsunami Early Warning System, 2016, http://www.ggos-portal.org/lang_en/GGOS-Portal/EN/Topics/ScienceApplications/DisasterMonitoring/TsunamiEarlyWarningSystem/TsunamiEarlyWarningSystem.html, accessed on 24 April 2016). The initiative is to act as a catalyst and a motivating force in development of an improved tsunami early warning capability for the Indo-Pacific Region. This vision is in keeping with the United Nations Global Geospatial Information Management (<http://ggim.un.org>) call for international cooperation on sharing geodetic data for development, science, and mitigation of natural hazards. An implementation of an Asia-Pacific tsunami warning augmentation system will require a combination of ground-based and spaceborne measurements and modeling capabilities as outlined in this paper.

6. Conclusions

In this paper, we have made the case that measurements of natural-hazards-generated ionospheric perturbations using ground-based and spaceborne observations combined with the use of sophisticated first-principles physics-based ionospheric modeling can be used to augment existing early warning systems, potentially saving human lives and minimizing economic damage.

Using ground-based and spaceborne GPS measurements, we demonstrated the observational evidence of atmosphere-ionosphere signatures associated with natural hazards including the 2011 Tohoku-Oki earthquake and tsunami. We found that the TEC perturbations are consistent with the earthquake-generated seismic waves. For the Tohoku-Oki event, the frequency band of TEC perturbations is in good agreement with the dominant periods of infrasound signals in the upper atmosphere (at the 445 km altitude of GRACE

such as COSMIC or GRACE data, constrains the WP-GITM model solution significantly, and uncertainties for the combined solution reduces to a 5 cm level tsunami wave height. This level of improvement in uncertainties is a major step forward in understanding the coupling between solid Earth, ocean, and atmosphere.

In Figure 5, we illustrate model simulations of surface displacements using GPS alone and combined GPS-and-spaceborne-data-derived estimates as a function of observed TEC perturba-

satellites at 06:07 UTC on 11 March 2011). Comparisons between ground infrasound waves and space-based observed neutral air density perturbations show a 9 min time delay, which can be interpreted as the infrasound propagation time from the ground to the GRACE altitude. JPL's atmosphere-ionosphere coupling simulations indicate a good agreement with the observations. The results provide additional arguments toward enhancing the GNSS ionosphere-based techniques for detection and imaging of natural hazards including earthquakes, tsunamis, and volcanic eruptions. In addition, the space-based remote sensing platforms (such as GRACE and the Gravity Field and Steady-State Ocean Circulation Explorer spacecraft) are expected to provide even more accurate observations that could be applied to validate the coupling between TIDs and acoustic gravity waves.

Using JPL's WP-GITM model simulations, we have come to the conclusion that space-based GPS measurements will decrease the uncertainties of estimated tsunami wave heights making the combination of ground-based and space-based measurements highly beneficial for future tsunami warning systems as more radio occultation satellites become operational in the near future including the upcoming COSMIC-2 constellation with highly accurate and near-real-time ionospheric radio occultation measurements. Initial simulation results suggest that we are able to retrieve tsunami wave heights and ground surface displacements at the 5 cm level of precision. Retrieving crucial geophysical parameters including, e.g., tsunami wave heights using TEC observations is paramount for implementing an Asia-Pacific tsunami warning augmentation system as initiated by GGOS.

Increasing the temporal and spatial coverage over the oceans using a combination of ground-based and spaceborne ionospheric measurements is expected to improve our understanding of earthquakes and subsequent tsunamis. The combined observations will make it possible to quantify energies for smaller and therefore more frequent events. We feel that it is imperative to understand and quantify the wave coupling of earthquakes and tsunamis to the ionosphere. We also aim at gaining a better understanding of errors affecting the estimated tsunami wave heights and vertical surface displacements that have direct impact on the coupling mechanism of ocean surface and atmospheric disturbances to the ionosphere.

Furthermore, ionospheric monitoring of TEC perturbations associated with earthquakes and tsunamis will improve scientists' understanding of postseismic ionospheric signatures in detail. The real-time monitoring of such signatures using both ground-based and spaceborne data is expected to become an integral part of the increasingly holistic approach such as GATEW. These systems will provide decision makers with critical information including the position, wavelength, and even the amplitude and scale of tsunamis as they move across the world's oceans.

Acknowledgments

The authors would like to thank NASA Headquarters, the Earth Science and Interior NASA ROSES grant program (NNH07ZDA001N-ESI) and the NASA Postdoctoral Program (NPP) fellowship administrated by Oak Ridge Associated Universities. R.B.L.'s involvement in the research is supported by the Natural Sciences and Engineering Research Council of Canada and the Canadian Space Agency. This research was performed at the Jet Propulsion Laboratory, California Institute of Technology, under contract to the National Aeronautics and Space Administration. The Japanese GEONET GPS data used in the paper may be requested from GNSS Earth Observatory Network System (GEONET Data, 2016, http://datahouse1.gsi.go.jp/terras/terra-s_english.html, accessed on 4 May 2016), and GRACE data are publicly available at COSMIC Data Analysis and Archive Center (CDAAC, online report, 2016). We gratefully acknowledge the U.S. Institute of Navigation for supplying Figures 2 and 4. The paper is based on the presentation originally given at the 2015 Ionospheric Effects Symposium in Alexandria, Virginia. Copyright 2016. All rights reserved.

References

- Arrowsmith, S. J., R. Burlacu, K. Pankow, B. Stump, R. Stead, R. Whitaker, and C. Hayward (2012), A seismoacoustic study of the 2011 January 3 Circleville earthquake, *Geophys. J. Int.*, **189**, 1148–1158, doi:10.1111/j.1365-246X.2012.05420.x.
- Artru, J., P. Lognonné, and E. Blanc (2001), Normal modes modeling of post-seismic ionospheric oscillations, *Geophys. Res. Lett.*, **28**, 697–700, doi:10.1029/2000GL000085.
- Artru, J., V. Ducic, H. Kanamori, P. Lognonné, and M. Murakami (2005a), Ionospheric detection of gravity waves induced by tsunamis, *Geophys. J. Int.*, **160**, 840–848, doi:10.1111/j.1365-246X.2005.02552.x.
- Artru, J., P. Lognonné, G. Occhipinti, F. Crespon, R. Garcia, E. Jeansou, and M. Murakami (2005b), Tsunami detection in the ionosphere, *Space Res. Today*, **163**, 23–27.
- Astafyeva, E., and K. Heki (2009), Dependence of waveform of near-field co-seismic-ionospheric disturbances on focal mechanisms, *Earth Planets Space*, **61**(7), 939–943.
- Astafyeva, E., K. Heki, E. Afraimovich, V. Kiryushkin, and S. Shalimov (2009), Two-mode long-distance propagation of coseismic ionosphere disturbances, *J. Geophys. Res.*, **114**, A10307, doi:10.1029/2008JA013853.
- Astafyeva, E., P. Lognonné, and L. M. Rolland (2011), First ionosphere images for the seismic slip on the example of the Tohoku-Oki earthquake, *Geophys. Res. Lett.*, **38**, L22104, doi:10.1029/2011GL049623.
- Astafyeva, E., S. Shalimov, E. Olshanskaya, and P. Lognonné (2013a), Ionospheric response to earthquakes of different magnitudes: Larger quakes perturb the ionosphere stronger and longer, *Geophys. Res. Lett.*, **40**, 1675–1681, doi:10.1002/grl.50398.
- Astafyeva, E., L. M. Rolland, P. Lognonné, K. Heki, and T. Yahagi (2013b), Parameters of seismic source as deduced from 1 Hz ionospheric GPS data: Case-study of the 2011 Tohoku-Oki event, *J. Geophys. Res. Space Physics*, **118**, 5942–5950, doi:10.1002/JGRA50556.
- Astafyeva, E., L. M. Rolland, and A. Sladen (2014), Strike-slip earthquakes can be detected in the ionosphere, *Earth Planet. Sci. Lett.*, **450**, 180–193.
- Bergeot, N., M. N. Bouin, M. Diamant, B. Pelletier, M. Régnier, S. Calmant, and V. Ballu (2009), Horizontal and vertical interseismic velocity fields in the Vanuatu subduction zone from GPS measurements: Evidence for a central Vanuatu locked zone, *J. Geophys. Res.*, **114**, B06405, doi:10.1029/2007JB005249.
- Calais, E., and J. B. Minster (1995), GPS detection of ionospheric perturbations following the January 17, 1994, Northridge earthquake, *Geophys. Res. Lett.*, **22**, 1045–1048, doi:10.1029/95GL00168.

- Calais, E., and J. B. Minster (1998), GPS, earthquakes, the ionosphere, and the Space Shuttle, *Phys. Earth Planet. Inter.*, **105**(3–4), 167–181.
- Coisson, P., P. Lognonné, D. Walwer, and L. M. Rolland (2015), First tsunami gravity wave detection in ionospheric radio occultation data, *Earth Space Sci.*, **2**, 125–133, doi:10.1002/2014EA000054.
- Crowell, B. W., D. Melgar, Y. Bock, J. S. Haase, and J. Geng (2013), Earthquake magnitude scaling using seismogeodetic data, *Geophys. Res. Lett.*, **40**, 6089–6094, doi:10.1002/2013GL058391.
- Deep-ocean Assessment and Reporting of Tsunamis (2016), DART buoy system information. [Available at http://nctr.pmel.noaa.gov/Dart/dart_home.html, Accessed on 23 April.]
- Dautermann, T., E. Calais, and G. S. Mattioli (2009a), Global Positioning System detection and energy estimation of the ionospheric wave caused by the 13 July 2003 explosion of the Soufrière Hills Volcano, Montserrat, *J. Geophys. Res.*, **114**, B02202, doi:10.1029/2008JB005722.
- Dautermann, T., E. Calais, P. Lognonné, and G. S. Mattioli (2009b), Lithosphere–atmosphere–ionosphere coupling after the 2003 explosive eruption of the Soufrière Hills Volcano, Montserrat, *Geophys. J. Int.*, **179**, 1537–1546, doi:10.1111/j.1365-246X.2009.04390.x.
- Ducic, V., J. Artru, and P. Lognonné (2003), Ionospheric remote sensing of the Denali Earthquake Rayleigh surface waves, *Geophys. Res. Lett.*, **30**(18), 1951, doi:10.1029/2003GL017812.
- e-POP/CASSIOPE (2016), Enhanced Polar Outflow Probe. [Available at <http://epop.phys.ucalgary.ca/quickfacts.html>, Accessed on 23 April.]
- Galvan, D. A., A. Komjathy, M. P. Hickey, and A. J. Mannucci (2011), The 2009 Samoa and 2010 Chile tsunamis as observed in the ionosphere using GPS total electron content, *J. Geophys. Res.*, **116**, A06318, doi:10.1029/2010JA016204.
- Galvan, D. A., A. Komjathy, M. Hickey, P. Stephens, J. B. Snively, T. Song, M. Butala, and A. J. Mannucci (2012), Ionospheric signatures of Tohoku-Oki tsunami of March 11, 2011: Model comparisons near the epicenter, *Radio Sci.*, **47**, RS4003, doi:10.1029/2012RS005023.
- Georges, T., and W. Hooke (1970), Wave-induced fluctuations in ionospheric electron content: A model indicating some observational biases, *J. Geophys. Res.*, **75**, 6295–6308, doi:10.1029/JA075i031p06295.
- Heale, C. J., and J. B. Snively (2015), Gravity wave propagation through a vertically and horizontally inhomogeneous background wind, *J. Geophys. Res. Atmos.*, **120**, 5931–5950, doi:10.1002/2015JD023505.
- Heki, K. (2006), Explosion energy of the 2004 eruption of the Asama Volcano, central Japan, inferred from ionospheric disturbances, *Geophys. Res. Lett.*, **33**, L14303, doi:10.1029/2006GL026249.
- Heki, K. (2011), Ionospheric electron enhancement preceding the 2011 Tohoku-Oki earthquake, *Geophys. Res. Lett.*, **38**, L17312, doi:10.1029/2011GL047908.
- Hickey, M. P., G. Schubert, and R. L. Walterscheid (2009), Propagation of tsunami-driven gravity waves into the thermosphere and ionosphere, *J. Geophys. Res.*, **114**, A08304, doi:10.1029/2009JA014105.
- Hickey, M. P., R. L. Walterscheid, and G. Schubert (2015), A full-wave model for a binary gas thermosphere: Effects of thermal conductivity and viscosity, *J. Geophys. Res. Space Physics*, **120**, 3074–3083, doi:10.1002/2014JA020583.
- Jin, S., G. Occhipinti, and R. Jin (2015), GNSS ionospheric seismology: Recent observation evidences and characteristics, *Earth Sci. Rev.*, **147**, 54–64, doi:10.1016/j.earscirev.2015.05.003.
- Jonah, O. F., E. A. Kherani, and E. R. De Paula (2016), Observation of TEC perturbation associated with medium-scale traveling ionospheric disturbance and possible seeding mechanism of atmospheric gravity wave at a Brazilian sector, *J. Geophys. Res. Space Physics*, **121**, 2531–2546, doi:10.1002/2015JA022273.
- Kanamori, H. (1977), The energy release in great earthquakes, *J. Geophys. Res.*, **82**(20), 2981–2987, doi:10.1029/JB082i020p0298.
- Kelley, M. C., R. Livingston, and M. McCready (1985), Large amplitude thermospheric oscillations induced by an earthquake, *Geophys. Res. Lett.*, **12**, 577–580, doi:10.1029/GL012i009p00577.
- Komjathy, A. (1997), Global ionospheric total electron content mapping using the Global Positioning System PhD dissertation, 248 pp., Department of Geodesy and Geomatics Eng. Tech. Rep. No. 188, Univ. of New Brunswick, Fredericton, New Brunswick, Canada.
- Komjathy, A. (2014), The Effects of the Ionosphere on the Propagation of Electromagnetic Waves, in *Encyclopedia of Remote Sensing*, edited by E. G. Njoku, Springer, New York.
- Komjathy, A., D. A. Galvan, P. Stephens, M. D. Butala, V. Akopian, B. D. Wilson, O. Verkhoglyadova, A. J. Mannucci, and M. Hickey (2012), Detecting ionospheric TEC perturbations caused by natural hazards using a global network of GPS receivers: The Tohoku case study, *Earth Planets Space*, **64**, 1287–1294, doi:10.5047/eps.2012.08.003.
- Komjathy, A., T. Song, and A. Buis (2013a), Drop in the ocean: Data from the global differential GPS network can predict the size of tsunami, *Meteorol. Technol. Int.*, 20–21.
- Komjathy, A., Y.-M. Yang, M. Butala, B. Iijima, and A. J. Mannucci (2013b), Detecting ionospheric TEC perturbations generated by natural hazards using a global network of GPS receiver In the Proceedings of the 2013 Beacon Satellite Symposium, Univ. of Bath, July 8–12, 2013.
- Komjathy, A., A. Romero-Wolf, Y.-M. Yang, R. B. Langley, and J. H. Foster (2014), CubeSat for Natural-Hazard Estimation With Ionospheric Sciences (CNEWS): A concept development to aid tsunami early warning systems, Abstract NH31C-3868 presented at 2014 Fall Meeting, AGU, San Francisco, Calif., 15–19 Dec.
- Komjathy, A., Y.-M. Yang, X. Meng, O. Verkhoglyadova, A. J. Mannucci, and R. B. Langley (2015a), Recent developments in understanding natural-hazards-generated TEC perturbations: Measurements and modeling results In the Proceedings of the 2015 Ionospheric Effects Symposium, Alexandria, Va, May 12–14 (Best Paper Award).
- Komjathy, A., Y.-M. Yang, X. Meng, O. Verkhoglyadova, A. J. Mannucci, and R. B. Langley (2015b), Detection of natural-hazards-generated TEC perturbations using ground-based and spaceborne ionospheric measurements and potential new applications In the Proceedings of the ION PNT Meeting, Honolulu, Hawaii, Apr 20–23.
- Lay, E. H., X.-M. Shao, and C. S. Carrano (2013), Variation in total electron content above large thunderstorms, *Geophys. Res. Lett.*, **40**, 1945–1949, doi:10.1002/grl.50499.
- Lognonné, P., J. Artru, R. Garcia, F. Crespon, V. Ducic, E. Jeansou, G. Occhipinti, J. Helbert, G. Moreaux, and P.-E. Godet (2006a), Ground based GPS tomography of ionospheric post-seismic signal, *Planet Space Sci.*, **54**, 528–540.
- Lognonné, P., R. Garcia, F. Crespon, G. Occhipinti, A. Kherani, and J. Artru (2006b), Seismic waves in the ionosphere, *Europhys. News*, **37**(4), 11–14.
- Mai, C.-L., and J.-F. Kiang (2009), Modeling of ionospheric perturbation by 2004 Sumatra tsunami, *Radio Sci.*, **44**, RS3011, doi:10.1029/2008RS004060.
- Makela, J. J., et al. (2011), Imaging and modeling the ionospheric airglow response over Hawaii to the tsunami generated by the Tohoku earthquake of 11 March 2011, *Geophys. Res. Lett.*, **38**, L00G02, doi:10.1029/2011GL047860.
- Mannucci, A. J., B. D. Wilson, D. N. Yuan, C. H. Ho, U. J. Lindqwister, and T. F. Runge (1998), A global mapping technique for GPS-derived ionospheric total electron content measurements, *Radio Sci.*, **33**, 565–582, doi:10.1029/97RS02707.
- Mazzotti, S., T. S. James, J. Henton, and J. Adams (2005), GPS crustal strain, postglacial rebound, and seismic hazard in eastern North America: The Saint Lawrence valley example, *J. Geophys. Res.*, **110**, B11301, doi:10.1029/2004JB003590.

- Melgar, D., and Y. Bock (2013), Near-field tsunami models with rapid earthquake source inversions from land and ocean-based observations: The potential for forecast and warning, *J. Geophys. Res. Solid Earth*, **118**, 5939–5955, doi:10.1002/2013JB010506.
- Melgar, D., and Y. Bock (2015), Kinematic earthquake source inversion and tsunami runup prediction with regional geophysical data, *J. Geophys. Res. Solid Earth*, **120**, 3324–3349, doi:10.1002/2014JB011832.
- Melgar, D., B. W. Crowell, Y. Bock, and J. S. Haase (2013), Rapid modeling of the 2011 M_w 9.0 Tohoku-Oki earthquake with seismogeodesy, *Geophys. Res. Lett.*, **40**, 2963–2968, doi:10.1002/grl.50590.
- Meng, X., A. Komjathy, O. P. Verkhoglyadova, Y.-M. Yang, Y. Deng, and A. J. Mannucci (2015), A new physics-based modeling approach for tsunami-ionosphere coupling, *Geophys. Res. Lett.*, **42**, doi:10.1002/2015GL064610.
- Nakashima, Y., K. Heki, A. Takeo, M. N. Cahyadi, and A. Aditiya (2014), Ionospheric disturbances by volcanic eruptions by GNSS-TEC: Comparison between Vulcanian and Plinian eruptions, Abstract NH33C-05 presented at 2014 Fall Meeting, AGU, San Francisco, Calif., 9–14 Dec.
- Occhipinti, G., P. Lognonné, E. A. Kherani, and H. Hebert (2006), Three-dimensional waveform modeling of ionospheric signature induced by the 2004 Sumatra tsunami, *Geophys. Res. Lett.*, **33**, L20104, doi:10.1029/2006GL026865.
- Occhipinti, G., E. A. Kherani, and P. Lognonné (2008), Geomagnetic dependence of ionospheric disturbances induced by tsunamigenic internal gravity waves, *Geophys. J. Int.*, **173**, 753–765, doi:10.1111/j.1365-246X.2008.03760.x.
- Occhipinti, G., P. Coisson, J. Makela, S. Allgeyer, A. Kherani, H. Hébert, and P. Lognonné (2011), Three-dimensional numerical modeling of tsunami-related internal gravity waves in the Hawaiian atmosphere, *Earth Planets Space*, **63**(7), 847–851, doi:10.5047/eps.2011.06.051.
- Occhipinti, G., L. M. Rolland, P. Lognonné, and S. Watada (2013), From Sumatra 2004 to Tohoku-Oki 2011: The systematic GPS detection of the ionospheric signature induced by tsunamigenic earthquakes, *J. Geophys. Res. Space Physics*, **118**, 3626–3636, doi:10.1002/jgra.50322.
- Peltier, W. R., and C. O. Hines (1976), On the possible detection of tsunamis by a monitoring of the ionosphere, *J. Geophys. Res.*, **81**(12), 1995–2000, doi:10.1029/JC081i012p01995.
- Physical Oceanography Distributed Archive Center (2016), GRACE data archive. [Available at <ftp://podaac-ftp.jpl.nasa.gov/allData/grace/L1B/JPL/RL02/>, Accessed on 23 April.]
- Rakoro, V., P. Lognonné, and L. Rolland (2015), Inversion of tsunami height using ionospheric observations. The case of the 2012 Haida Gwaii tsunami, Abstract NH31C-3878 presented at 2015 Fall Meeting, AGU, San Francisco, Calif., Dec.
- Ridley, A. J., Y. Deng, and G. Toth (2006), The global ionosphere-thermosphere model, *J. Atmos. Sol. Terr. Phys.*, **68**, 839–864, doi:10.1016/j.jastp.2006.01.008.
- Rolland, L. M., P. Lognonné, and H. Munkane (2011), Detection and modeling of Rayleigh wave induced patterns in the ionosphere, *J. Geophys. Res.*, **116**, A05320, doi:10.1029/2010JA016060.
- Rolland, L. M., M. Vergnolle, J.-M. Nocquet, A. Sladen, J.-X. Dessa, F. Tavakoli, H. R. Nankali, and F. Cappa (2013), Discriminating the tectonic and non-tectonic contributions in the ionospheric signature of the 2011, M_w 7.1, dip-slip Van earthquake, Eastern Turkey, *Geophys. Res. Lett.*, **40**, 2518–2522, doi:10.1002/grl.50544.
- Roy, C., G. Occhipinti, L. Boschi, J.-P. Molin, and M. Wieczorek (2014), Effect of ray and speed perturbations on ionospheric tomography by over-the-horizon radar: A new method, *J. Geophys. Res. Space Physics*, **119**, 7841–7857, doi:10.1002/2014JA020137.
- Simons, M., et al. (2011), The 2011 magnitude 9.0 Tohoku-Oki earthquake: Mosaicking the megathrust from seconds to centuries, *Science*, **332**(6036), 1421–1425, doi:10.1126/science.1206731.
- Snively, J. B. (2013), Mesospheric hydroxyl airglow signatures of acoustic and gravity waves generated by transient tropospheric forcing, *Geophys. Res. Lett.*, **40**, 4533–4537, doi:10.1002/grl.50886.
- Song, Y. T. (2007), Detecting tsunami genesis and scales directly from coastal GPS stations, *Geophys. Res. Lett.*, **34**, L19602, doi:10.1029/2007GL031681.
- Song, Y. T., and S. C. Han (2011), Satellite observations defying the long-held tsunami genesis theory, in *Remote Sensing of the Changing Oceans*, edited by D. L. Tang, Springer, Berlin, doi:10.1007/978-3-642-16541-2.
- Song, Y. T., L.-L. Fu, V. Zlotnicki, C. Ji, V. Hjordlefsdottir, C. K. Shum, and Y. Yi (2008), The role of horizontal impulses of the faulting continental slope in generating the 26 December 2004 tsunami, *Ocean Model.*, doi:10.1016/j.ocemod.2007.10.007.
- Song, Y. T., I. Fukumori, C. K. Shum, and Y. Yi (2012), Merging tsunamis of the 2011 Tohoku-Oki earthquake detected over the open ocean, *Geophys. Res. Lett.*, **39**, L05606, doi:10.1029/2011GL050767.
- Stephens, P., A. Komjathy, B. Wilson, and A. Mannucci (2011), New leveling and bias estimation algorithms for processing COSMIC/FORMOSAT-3 data for slant total electron content measurements, *Radio Sci.*, **46**, RS0D10, doi:10.1029/2010RS004588.
- Tang, L., et al. (2012), Direct energy estimation of the 2011 Japan tsunami using deep-ocean pressure measurements, *J. Geophys. Res.*, **117**, C08008, doi:10.1029/2011JC007635.
- Titov, V. V., and F. I. Gonzales (1997), Implementation and testing of the Method of Splitting Tsunami (MOST) model, NOAA Tech. Memo., ERL PMEL112, 11 pp. [Available at <http://www.pmel.noaa.gov/pubs/PDF/tito1927/tito1927.pdf>.]
- Titov, V. V., C. W. Moore, D. J. M. Greenslade, C. Pattiaratchi, R. Badal, C. E. Synolakis, and U. Kanoğlu (2011), A new tool for inundation modeling: Community Modeling Interface for Tsunamis (ComMIT), *Pure Appl. Geophys.*, **168**, 2121–2131, doi:10.1007/s00024-011-0292-4.
- United States Geological Survey (2016), [Available at <http://earthquake.usgs.gov/earthquakes/>, Accessed on 23 April.]
- Vadas, S. L., and M. J. Nicolls (2012), The phases and amplitudes of gravity waves propagating and dissipating in the thermosphere: Theory, *J. Geophys. Res.*, **117**, A05322, doi:10.1029/2011JA017426.
- Watada, S., and H. Kanamori (2010), Acoustic resonant oscillations between the atmosphere and the solid earth during the 1991 Mt. Pinatubo eruption, *J. Geophys. Res.*, **115**, B12319, doi:10.1029/2010JB007747.
- Yang, Y.-M., J. L. Garrison, and S.-C. Lee (2012), Ionospheric disturbances observed coincident with the 2006 and 2009 North Korean underground nuclear tests, *Geophys. Res. Lett.*, **39**, L02103, doi:10.1029/2011GL050428.
- Yang, Y.-M., A. Komjathy, R. B. Langley, P. Vergados, M. D. Butala, and A. J. Mannucci (2014a), The 2013 Chelyabinsk meteor ionospheric impact studied using GPS measurements, *Radio Sci.*, **49**, 341–350, doi:10.1002/2013RS005344.
- Yang, Y.-M., X. Meng, A. Komjathy, O. Verkholyadova, R. B. Langley, B. T. Tsurutani, and A. J. Mannucci (2014b), Tohoku-Oki earthquake caused major ionospheric disturbances at 450 km altitude over Alaska, *Radio Sci.*, **49**, 1206–1213, doi:10.1002/2014RS005580.
- Yuen, P. C., P. F. Weaver, and R. K. Suzuki (1969), Continuous, traveling coupling between seismic waves and the ionosphere evident in May 1968 Japan earthquake data, *J. Geophys. Res.*, **74**, 2256–2264, doi:10.1029/JA074i009p02256.
- Zabotin, N. A., O. A. Godin, and T. W. Bullett (2016), Oceans are a major source of waves in the thermosphere, *J. Geophys. Res. Space Physics*, **121**, doi:10.1002/2016JA022357.
- Zettergren, M., and J. B. Snively (2015), Ionospheric response to infrasonic-acoustic waves generated by natural hazard events, *J. Geophys. Res. Space Physics*, **120**, 8002–8024, doi:10.1002/2015JA021116.